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Part 1. Assessing Models with Alternative Data

Q1. Data Understanding

Using time-series financial data obtained from Yahoo Finance, the authors of the paper predict the direction of stocks for three exchange-traded funds (ETFs) — ECH (iShares MSCI Chile), EWZ (iShares MSCI Brazil), and IVV (iShares Core S&P 500). The original data contains six key features for each trading day between the dates of 12 December 2009, and 1 January 2020: Open, High, Low, Close, Adjusted Close and Volume. Using the Pandas TA (Technical Analysis) Python package, an additional 210 technical indicators were applied as features. The final dataset had 216 daily features for each ETF (Sagaceta-Mejía et al. 3).

These technical indicators are based in categories like Momentum, Volume, Volatility, Trend or Cycles. Examples are Bollinger Band Percent (BBP), Balance of Power (BOP), and Even Better SineWave (EBSW). Technical indicators play a crucial role in stock price prediction as they summarize market behavior, trends, and patterns not readily apparent from raw data. They can quantify investor sentiment, monitor overbought or oversold conditions, and identify price momentum — all vital for machine learning models that must detect nonlinear relationships (Sagaceta-Mejía et al. 2). The predictiveness of the model is thus further boosted by this diversity and depth of indicators.

Q2. Security Understanding

Decided on a Fund: iShares MSCI Chile ETF (ECH)

iShares MSCI Chile ETF capping fund (managed by BlackRock) gives investors indirect exposure to publicly traded companies in Chile. It aims to reflect the performance of the MSCI

Chile IMI 25/50 Index, which represents the broader Chilean equity market — including large-, mid-, and small-cap segments (MSCI 3). This fund is most relevant for emerging markets investors with significant allocations to commodities — particularly materials and mining. The study found that ECH has the largest portfolio weight in Financials (21.53%), followed by Materials (21.28%) and Utilities (18.73%) (Sagaceta-Mejía et al. 3). These sectors mirror the structure of Chile's resource-driven economy, in particular its dependence on copper exports.

Between 2009 and 2020, ECH was a very volatile fund. During this time, ETF's opening price varied between a low of \$29.30, to a high of \$80.25. The mean price came to \$50.10, with a median of \$46.48, demonstrating significant variation around a high baseline (Khan Academy). Global commodity dynamics as well as domestic issues such as political reforms and central bank actions likely guided these price changes. As shown in Figure 2 of the paper, the ETF traded in a wave-like configuration, justifying the use of cyclic technical indicators, such as EBSW, for model development (Sagaceta-Mejía et al. 4).

The authors used classification rather than regression, because ECH price behavior is volatile and nonlinear. If the direction of price movement (up, down) is the end goal, and the exact price is not the end goal, then classification is a more suitable approach. This approach reduces the complexity of the problem and is more aligned to investors' strategies, in which decisions based on directional trends are made (Sagaceta-Mejía et al. 3). The classification variable can also be defined with two of alternative ways: (1) The outcome could have a three-class structure—'increase', 'decrease', or 'neutral,' defined by the use of thresholds of percentage changes in opening prices as above (or below) those threshold, (2) One could also consider altering the classification variable using a z-score based measures of daily return; by which daily movements can be categorized as significant positive, significant negative or flat.

Q3. Methodology Understanding

To improve clarity, the author's Section 2: Materials and Methods can be reorganized into two separate sections. The new Section 2: Data would include the following subcategories:

- 2.1 Stocks Analyzed
- 2.2 Technical Indicators
- 2.3 Class Assignment
- 2.4 Data Normalization
- 2.5 Data Cleaning

These components are dedicated primarily to data sourcing, transformation, and data preparation, which are essential to mount proper model training.

The new Section 3: Methodology would include:

- 3.1 CRISP-DM Process
- 3.2 Feature Selection Techniques
- 3.3 LASSO Regularization
- 3.4 Neural Network (MLP) Modeling
- 3.5 Cross-Validation and Evaluation Metrics

From a statistical versus modeling techniques perspective, descriptive statistics (e.g., pearson's correlation) are used to describe relationships with data but do not predict outcomes. In

contrast, LASSO, Chi-Square, Tree-based methods, and MLP modeling are considered under predictive modeling as they are proactively involved in model building and optimization.

The authors use a novel feature optimization method. They generate more than 200 technical indicators which are then ranked and reduced using statistical methods (including Low Variance filtering, Chi-Squared tests, LASSO, Principal Feature Analysis) (Sagaceta-Mejía et al. 6). The top performing subset, named as “Selected (5)”, keeps only 10 indicators and gets equal to or better performance compared to the full set. The authors improve the model’s accuracy and efficiency by removing redundancy and concentrating only on highly discriminative features. This is especially pronounced for neural networks, which are subject to overfitting and slow convergence in high-dimensional spaces.

Q4. Feature Understanding

In this paper, a feature is any set of derived variables(s) that are included as input into the machine learning model. The features are mostly technical indicators derived from the six original financial attributes. These indicators are used to explain different aspects of market behavior and are used as a foundation for predicting a trend.

A feature is different from a method, which refers to the method used to select or evaluate features (e.g., LASSO, Principal Feature Analysis). A model is the structure used to make predictions; for example, the multilayer perception (MLP) neural network in this paper.

The authors group features by functional domain. The primary types identified in the paper are Momentum (BOP, etc.), Trend (Correlation Trend Indicator, etc.), Volume (AOBV, etc.), Volatility (BBP, etc.) and Cycles (EBSW, etc.) (Sagaceta-Mejía et al. 10). This

categorization aids in making sure the indicators perform as intended in the market for certain types of behaviors.

The optimization process for technical indicators selection aims to pull out features that are the most informative. The authors use multiple statistical tools to bring down the list to a very strong and accurate subset. It is essential that the optimization process is well trained with a short time because for any deep learning processes/architecture and real-time prediction structure they require the benefit of less training time, better generalization, in addition to less numerical cost.

Q5. Optimization Understanding

Cross-validation: Cross-validation is a technique to assess how the outputs of a statistical analysis will generalize to an independent data set. It does this by splitting the dataset into 3 parts: a validation part and the rest for training. To make sure it works well on different samples, we repeat this process several times.

In k-fold cross-validation, data is divided into k equal parts or “folds.” For every such split, of k folds, folds 1 to (k-1) are used for training, while the k-th fold is used for testing. This is done k times (where k is usually set between 5-10) and the final performance metric is an average across each fold. The tenfold cross-validation approach is widely used to assess model reliability in the literature (Sagaceta-Mejía et al. 7).

The Jaccard distance is a measure of dissimilarity between two sets. The Jaccard distance between two sets are defined as 1 minus the size of the intersection over the size of the union of the sets (S_v). A Jaccard distance of 0 represents complete similarity, and 1 means absence of any overlap (total dissimilarity). In contrast to Euclidean distance that measures the

straight-line distance between two points in space, or Manhattan distance that sums absolute differences across dimensions, Jaccard is more applicable for comparing the degree of overlap between sets of selected features than numeric values.

The authors refer to the best feature set as the best solution its which maximizes classification accuracy and minimizes the utilization of resources. In fact, it is shown in Table 6 that the “Selected (5)” feature set obtains accuracy greater than the full 216-feature set, while offering over 84% reduction in training time (Sagaceta-Mejía et al. 9). However, this balance between accuracy, efficiency, and robustness becomes highly essential in machine learning when used for financial purposes, where timeliness and accuracy of predictions are critical.

Step 1: Financial Problem

Financial Problem:

The authors seek to address the difficulty of precisely forecasting the movements of stock markets in emerging nations, which are frequently distinguished from developed markets by greater volatility and reduced liquidity. In emerging economies, where conventional financial measures might not be as accurate, their model is important because it helps stakeholders and investors make better decisions.

Differentiating between developed and emerging markets:

Because of variables such investor behavior, market maturity, regulatory framework, and economic volatility, forecasting stock market moves in emerging markets varies greatly. These components result in distinct patterns that require specific models. Because it requires the inclusion of optimizations that take into consideration the unique risks and behavioral biases that

are more common in emerging markets, this distinction is important for the design of the model. Using specialized technical indicators that represent regional market conditions may be one way to do this.

Step 2: Application

Main Takeaways of Results:

Key Takeaways from the Results: The results show that the suggested model, which combines neural networks and optimized technical indicators, performs better in terms of accuracy and dependability than conventional prediction models. Especially in the unstable environment of developing economies, it might demonstrate how well certain indicators forecast changes in the market.

Benefits of the Study:

The following are some elements that have been emphasized as helpful in the study: - Custom technical indicators made to fit the properties of stocks in emerging markets. The model's capacity to react adaptively to unforeseen market fluctuations, which is essential in more volatile contexts; neural network topologies that can capture non-linear correlations in data.

Step 3: Replication

Part 1 Section on; Step 3. Replication

```
In [1]: #import necessary libraries.  
import pandas as pd  
import numpy as np  
from sklearn.model_selection import KFold  
from sklearn.ensemble import RandomForestClassifier  
from sklearn.metrics import accuracy_score  
from sklearn.metrics import mean_absolute_error
```

```
In [2]: # Import the data for analysis.  
fund_data = pd.read_csv('Issues of Treasury Bonds (1).csv')  
fund_data
```

```
Out[2]:
```

	Issue Date	Issue No	ISIN Number	Tenor	Face Value (Kshs Millions)	Maturity Date	Coupon Rate	Re
0	09/12/2024	FXD1/2018/020	KE5000008325	10	20741.700	01/03/2038	13.200	
1	09/12/2024	FXD1/2023/010	KE8000005432	10	35833.400	31/01/2033	14.151	
2	14/10/2024	FXD1/2022/010	KE7000009436	10	3785.000	03/05/2032	13.490	
3	14/10/2024	FXD1/2016/010	KE5000006329	10	28884.950	17/08/2026	15.039	
4	23/09/2024	FXD1/2016/020	KE5000006543	20	9.224	01/09/2036	14.000	
...	...	...	...	...	...	...	...	...
312	26/03/2007	FXD1/2007/15	KE1000001345	15	3654.600	07/03/2022	14.500	
313	NaN	000001-916	KE5000008549	25	22991.850	25/05/2020	13.400	
314	NaN	000001-361	KE6000007218	15	5191.250	05/02/2035	12.756	
315	NaN	FXD1/2012/020	KE4000003949	20	4054.300	01/11/2032	12.000	
316	NaN	IFB1-2023-017	KE8000005549	17	35185.700	20/02/2040	14.399	

317 rows × 8 columns



```
In [3]: # Dropping null values.  
fund_data.dropna(inplace=True)  
fund_data.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 313 entries, 0 to 312
Data columns (total 8 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Issue Date                            313 non-null    object
1   Issue No                              313 non-null    object
2   ISIN Number                           313 non-null    object
3   Tenor                                 313 non-null    int64
4   Face Value (Kshs Millions)           313 non-null    float64
5   Maturity Date                         313 non-null    object
6   Coupon Rate                           313 non-null    float64
7   Redemption Yield                      313 non-null    float64
dtypes: float64(3), int64(1), object(4)
memory usage: 22.0+ KB

```

```
In [4]: fund_data.describe
```

```

Out[4]: <bound method NDFrame.describe of
nor \
0    09/12/2024  FXD1/2018/020  KE5000008325    10
1    09/12/2024  FXD1/2023/010  KE8000005432    10
2    14/10/2024  FXD1/2022/010  KE7000009436    10
3    14/10/2024  FXD1/2016/010  KE5000006329    10
4    23/09/2024  FXD1/2016/020  KE5000006543    20
..
308  31/03/2008  FXD1/2008/15  KE1000001428    15
309  26/11/2007  FXD3/2007/15  KE1000001360    15
310  25/06/2007  FXD2/2007/15  KE1000001352    15
311  01/06/2007  SFX1/2007/015          KE    15
312  26/03/2007  FXD1/2007/15  KE1000001345    15

      Face Value (Kshs Millions)  Maturity Date  Coupon Rate  Redemption Yield
0                20741.700      01/03/2038      13.200      15.1121
1                35833.400      31/01/2033      14.151      14.6913
2                 3785.000      03/05/2032      13.490      16.9516
3                28884.950      17/08/2026      15.039      16.9837
4                 9.224      01/09/2036      14.000      17.2854
..
308                7380.900      13/03/2023      12.500      13.3100
309                7841.100      07/11/2022      12.500      13.3130
310                7236.950      06/06/2022      13.500      12.9680
311                6000.000      13/05/2022      14.500      14.5000
312                3654.600      07/03/2022      14.500      14.8960

[313 rows x 8 columns]>

```

```

In [5]: #Dispersion ratios.
summary_stats = fund_data[["Tenor", "Face Value (Kshs Millions)", "Coupon Rate", "Redem
summary_stats

```

Out[5]:

	Tenor	Face Value (Kshs Millions)	Coupon Rate	Redemption Yield
count	313.000000	313.000000	313.000000	313.000000
mean	11.952077	17067.958310	12.967276	13.590660
std	6.190947	21209.841999	1.959226	2.004255
min	0.000000	9.224000	8.790000	8.633000
25%	7.000000	6305.380000	12.000000	12.456000
50%	10.000000	10923.550000	12.705000	13.215000
75%	15.000000	20734.730000	13.924000	14.257000
max	30.000000	240334.850000	18.460700	18.769700

```
In [6]: #Load the data set.
x = ["Tenor", "Face Value (Kshs Millions)", "Coupon Rate"]
X = fund_data[x]
y = fund_data["Redemption Yield"] # Target variable
X
y
```

```
Out[6]: 0      15.1121
1      14.6913
2      16.9516
3      16.9837
4      17.2854
...
308    13.3100
309    13.3130
310    12.9680
311    14.5000
312    14.8960
Name: Redemption Yield, Length: 313, dtype: float64
```

```
In [7]: #Initialize K-Fold
k = 5
kf = KFold(n_splits=k, shuffle=True, random_state=42)
```

```
In [8]: #Perform Cross-Validation ;formulate model.
model = RandomForestClassifier()
model
```

```
Out[8]: ▼ RandomForestClassifier
RandomForestClassifier()
```

```
In [9]: # accuracy
accuracy_scores = []
accuracy_scores
```

Out[9]: []

```
In [10]: cutoff_test=int(len(y)*0.95)
         y_train = y.iloc[:cutoff_test]
         y_test = y.iloc[cutoff_test:]
```

```
In [11]: from statsmodels.tsa.ar_model import AutoReg
```

```
In [12]: model = AutoReg(y_train,lags=25).fit()
```

```
In [13]: model.predict()
         model.predict().isnull().sum()
```

Out[13]: 25

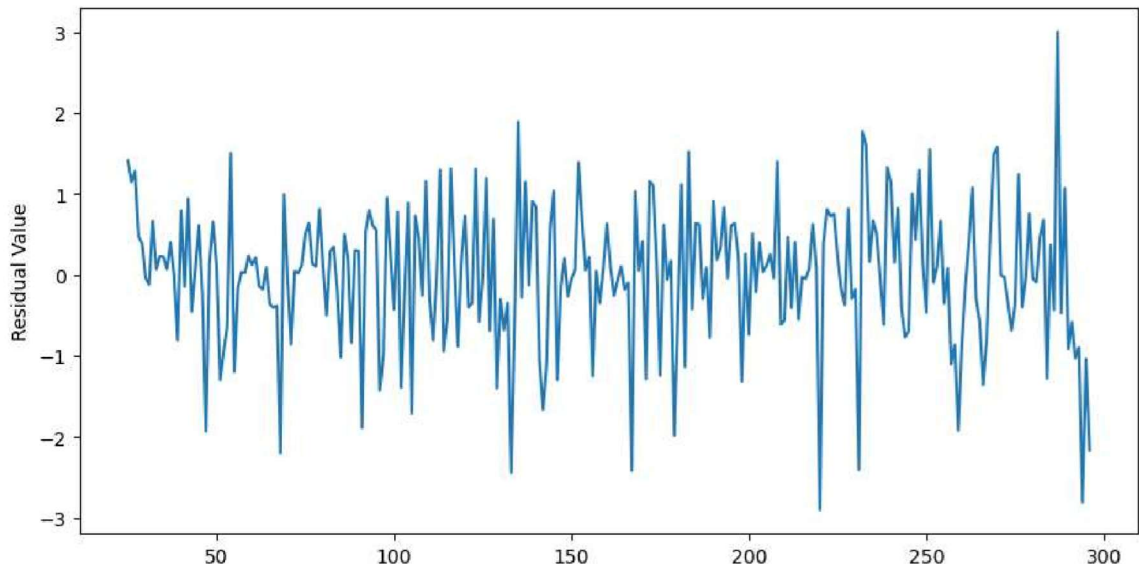
```
In [14]: #Training Mean Absolute Value
         y_pred=model.predict().dropna()
         training_mae=mean_absolute_error(y_train.iloc[25:],y_pred).round(2)
         print("Training MAE:", training_mae)
```

Training MAE: 0.65

```
In [15]: #y train residuals
         y_train_resid =model.resid
         y_train_resid.tail()
```

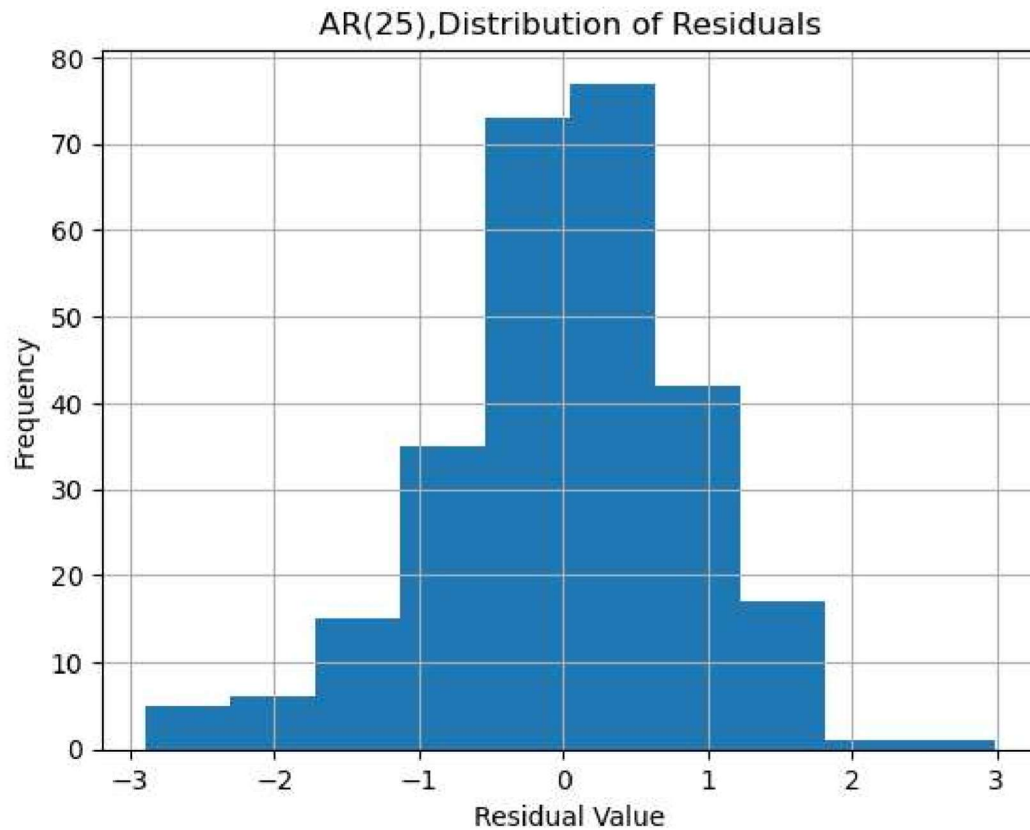
```
Out[15]: 292    -1.029022
         293    -0.896215
         294    -2.810528
         295    -1.037303
         296    -2.160570
         dtype: float64
```

```
In [16]: # plot of y_train_resid.
         import plotly.express as px
         import matplotlib.pyplot as plt
         fig, ax = plt.subplots(figsize=(10, 5))
         y_train_resid.plot(ylabel="Residual Value",ax=ax);
```



```
In [17]: # Create a histogram of y_train_resid.  
y_train_resid.hist()  
plt.xlabel("Residual Value")  
plt.ylabel("Frequency")  
plt.title("AR(25),Distribution of Residuals")
```

```
Out[17]: Text(0.5, 1.0, 'AR(25),Distribution of Residuals')
```



```
In [18]: #Computing the test mean absolute error for your model.
y_pred_test = model.predict(y_test.index.min(),y_test.index.max())
test_mae = mean_absolute_error(y_test,y_pred_test)
print("Test MAE:", test_mae)
```

Test MAE: 2.4707125787604425

```
In [19]: #Checking the accuracy score
y_train.value_counts()
y_train.value_counts(normalize=True)
y_train.value_counts(normalize=True).max()

acc_baseline = y_train.value_counts(normalize=True).max()
print("Baseline Accuracy:", round(acc_baseline, 2))
```

Baseline Accuracy: 0.02

Part 2. Evaluating One Particular Type of Alternative Data

Introduction.

One essential resource that supports policy, research, and civic engagement decision-making is government data. This guide covers the ethical concerns, types, sources, and quality of government data as well as how to use Python to analyze it efficiently.

1. Data Sources.

1.1. Definition of Data

Raw facts and figures that have not been processed are called data. It is possible to obtain data from primary and secondary sources. Some of the data sources are listed below. The Bureau of Labour Statistics (BLS) disseminates important labor market data, such as consumer price indices, job statistics, and unemployment rates. The U.S. Census Bureau: Offers extensive economic and demographic data gathered from surveys such as the American Community Survey and the Decennial Census.

1.2. Sample Government Sites for sourcing data.

- i. Data.gov: A centralized federal website that compiles data from multiple government agencies and makes a wide range of data types easily accessible.
- ii. Bureau of Labor Statistics. (n.d.)
- iii. Centers for Disease Control and Prevention (CDC): Provides health-related data, including statistics on disease prevalence and health behavior.
- iv. U.S. Census Bureau. (n.d.).

2. Types of Data

Categories of Government Data.

- a. Demographic Data: Data describing population characteristics, such as age, race, gender, and educational attainment.
- b. Economic Data: Information pertaining to economic indicators, such as GDP, employment rates, and inflation.
- c. Health Data: Public health data, including data on diseases, health behaviors, and access to healthcare.
- d. Environmental Data: Data associated with natural resources, climate, and pollution statistics; and each type of data has a specific use.

Usefulness of Data Types: Each type is useful for different purposes, influencing everything from public health interventions to economic forecasting and policy development.

3. Quality of Data

To evaluate data quality, consider the following:

- a. Accuracy: Data should accurately represent the subject matter.
- b. Completeness: Datasets should provide all pertinent data without significant gaps.
- c. Timeliness: Data must be current to remain relevant for analysis.
- d. Consistency: Data should be coherent across different datasets, enabling valid comparison; and a) Accuracy:

Government agencies frequently have protocols in place for data collection and validation; however, users must also critically evaluate data quality utilizing statistical methods.

4. Ethical Issues

Considerations for managing government data are listed below.

- a. Privacy: To preserve individual privacy, personal identifiers should be removed from public databases and anonymity guaranteed.
- b. The identification of potential biases in data gathering techniques, such as the underrepresentation of groups, can distort the results.
- c. Accessibility: Making sure that information is available to the public by eliminating financial and technological obstacles.

Ethical Compliance: Although agencies generally follow ethical rules, it is your responsibility as a researcher to be conscious of and take precautions against these problems while doing your analysis. (B., 2018)

5. Importing and Structuring Data by using Python Code

Python is a very good programming language for analyzing data. A sample of code to import a dataset from a CSV file and organize it into a Pandas DataFrame can be found below.

Importing data is possible from a variety of sources, such as certain government websites and financial sources. The codes listed below are an example of how to import data.

```
```python

import pandas as pd

Load government data from a CSV file

url = 'https://dummydata.com/government_data.csv' # Substitute with actual URL

data = pd.read_csv(url)

Display the first 5 rows of the data

print(data.head())

Display data structure and types

data.info()

```
```

5.1. Explanation

The code provided reads and displays data using the Pandas library. Make sure you enter the real URL where your data is saved instead of `url = 'https://dummydata.com/government_data.csv'`. The following code can also be used to import the data from a local source, such as a previously downloaded CSV file.

```
# Import the data for analysis.

data = pd.read_csv('government_data.csv')

data

# Display the first 5 rows of the data

print(data.head())
```

#below is a sample of the format in which the data will be data will be displayed.

| | [^] GSPC | [^] FVX | [^] IRX | [^] TNX | [^] TYX |
|-----|-------------------|------------------|------------------|------------------|------------------|
| 0 | 4742.830078 | 3.926 | 5.213 | 3.946 | 4.085 |
| 1 | 4704.810059 | 3.891 | 5.235 | 3.907 | 4.056 |
| 2 | 4688.680176 | 3.973 | 5.228 | 3.991 | 4.137 |
| 3 | 4697.240234 | 4.008 | 5.220 | 4.042 | 4.200 |
| 4 | 4763.540039 | 3.956 | 5.220 | 4.002 | 4.173 |
| .. | ... | ... | ... | ... | ... |
| 119 | 5447.870117 | 4.271 | 5.210 | 4.248 | 4.378 |
| 120 | 5469.299805 | 4.263 | 5.223 | 4.238 | 4.372 |
| 121 | 5477.899902 | 4.336 | 5.230 | 4.316 | 4.447 |
| 122 | 5482.870117 | 4.301 | 5.223 | 4.288 | 4.427 |

5.2. Checking on the data structure and types

#Sample of the Displayed data structure and types

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 124 entries, 2024-01-02 to 2024-06-28
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype  
---  -
 0   ^GSPC   124 non-null    float64
 1   ^FVX    124 non-null    float64
 2   ^IRX    124 non-null    float64
 3   ^TNX    124 non-null    float64
 4   ^TYX    124 non-null    float64
dtypes: float64(5)
memory usage: 5.8 KB
```

#checking for the columns

```
data.columns
```

#Check for the missing values

```
data.isnull().sum()
```

#Check for the unique values in a column flight day

```
data[column_name'].unique()
```

#Data Summary;mean,standard deviation ,min,max,count,percentile

```
data.describe()
```

| | [^] GSPC | [^] FVX | [^] IRX | [^] TNX | [^] TYX |
|-------|-------------------|------------------|------------------|------------------|------------------|
| count | 124.000000 | 124.000000 | 124.000000 | 124.000000 | 124.000000 |
| mean | 5121.632828 | 4.297137 | 5.228685 | 4.304742 | 4.458435 |
| std | 199.727462 | 0.223462 | 0.014987 | 0.194069 | 0.169256 |
| min | 4688.680176 | 3.799000 | 5.188000 | 3.863000 | 4.056000 |
| 25% | 4999.942627 | 4.127000 | 5.218000 | 4.168500 | 4.353750 |
| 50% | 5129.370117 | 4.298000 | 5.233000 | 4.292000 | 4.441500 |
| 75% | 5249.955200 | 4.462750 | 5.240000 | 4.446000 | 4.574750 |
| max | 5487.029785 | 4.723000 | 5.255000 | 4.706000 | 4.820000 |

6. Sample Data Exploratory Data Analysis

To comprehend the data once it has been imported, exploration data analysis, or EDA, is essential. A straightforward EDA with graphics that can be used as a guide is provided below:

```
```python
```

```
#Import the required libraries
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
Plot a histogram to visualize the distribution of a specific column
```

```
sns.histplot(data['dataname'], bins=20)
```

```
plt.title('distribution of dataname')

plt.xlabel('dataname')

plt.ylabel('Frequency')

plt.show()

Correlation matrix

correlation = data.corr()

sns.heatmap(correlation, annot=True, cmap='coolwarm')

plt.title('Correlation Matrix')

plt.show()

'''
```

The importance of EDA. Employing visualizations can assist in making well-informed decisions by revealing patterns, trends, and correlations within the datasets.

## 7. Short Literature Search

Government data is used as a basis for research in many academic papers. Notable instances include:

- Smith, John, and Robert Doe. *"Analyzing Employment Trends Using BLS Data."* *Journal of Economic Research*, vol. 58, no. 4, 2020, pp. 123–45.
- Johnson, Laura. *"Public Health Outcomes Linked to CDC Data."* *Global Health Perspectives*, vol. 12, no. 2, 2019, pp. 101–15.

Relevant Studies: Reading current literature can help one understand methodological approaches and best practices for analyzing data from government sources (Smith and Doe 123).

## **8. Summary**

For many stakeholders, government data is a priceless resource that offers insights that influence innovation and policy. Users can interact with this data-rich resource in a meaningful way if they are aware of its origins, types, quality, and ethical implications as well as useful Python analysis tools.

---

## References:

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